

February 27, 2025

ECEN-607: Advanced Analog IC Design Assignment #4

Due: March 17, 2026

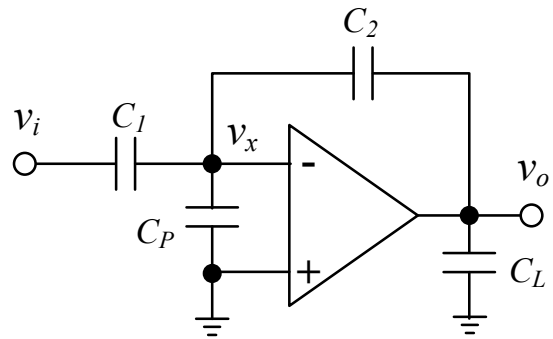
Submit your assignments through Canvas

Homework will not be received after due.

Instructor: Dr. Jose Silva-Martinez

Repeat your assignment 3, but use the following architecture: This design is fully-differential.
<https://ieeexplore.ieee.org/abstract/document/11165015/metrics#metrics>

- DC gain $> 50\text{dB}$
- Total power must be limited to 10mWatts; reduce as much as possible the time for slew and linear settling.
- $C_1=800\text{fF}$, $C_2=100\text{fF}$, $C_L=200\text{fF}$; C_P represents the parasitic capacitors of the amplifier and should not be added (remove C_P).
- The circuit must be fully-differential, then a common-mode feedback engine must be included.



- i) Measure the AC properties of the loop gain; do your best for the stabilization of the architecture. Report the magnitude and phase response and measure the DC gain and phase and gain margin.
- ii) Measure the 100mV step response. Report the steady state settling error and the 1% settling time.
- iii) Compare the results for this architecture and the ones obtained in assignment 3.

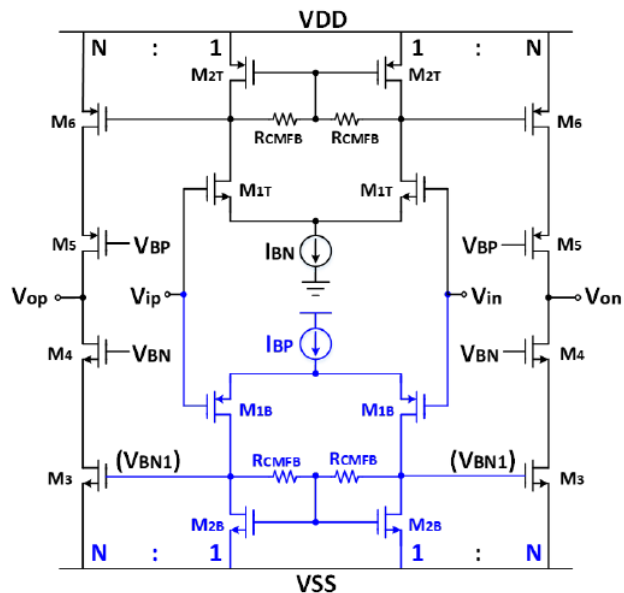


Fig. 3. Proposed dual inputs two stage amplifier architecture (D2SA)

Prepare a report of no more than 4 pages. **Your report must be concrete but clear; include relevant screen shots with enough resolution; font size of the axes must be 12 points or higher to make them visible!**